

ANTHONY CATALINA

Full-Stack & AI Engineer · React & TypeScript · Agentic Workflows

Fort Collins, CO · (609) 651-0935 · antcatalina1@gmail.com · antcatalina.com · linkedin.com/in/anthony-catalina

SUMMARY

Full-stack engineer with 5+ years delivering production React/TypeScript applications at enterprise scale, with hands-on experience embedding LLM agents (Claude Code, GitHub Copilot) into daily frontend engineering, including MCP-integrated design and documentation workflows and autonomous nightly agents with production guardrails. Unique background spanning embedded systems and modern UI development, with a strong track record building real-time dashboards, component libraries, and design systems for Fortune 500 clients across IoT, industrial, and distribution industries.

SKILLS

Frameworks & Languages: React, TypeScript, JavaScript, Lit, HTML5, CSS3/SCSS, Bootstrap, Python

Architecture & Real-Time Data: Atomic Design, Component-Driven Development, MVC, Monorepo, Design Systems, REST APIs, MQTT, WebSocket

DevOps & Tooling: AWS S3, GitHub Actions CI/CD, Jenkins, Git, Bitbucket, SureMDM

AI Engineering: Claude Code, GitHub Copilot, Anthropic API, MCP protocol and server integration (Figma, Atlassian/Confluence, Jira), agentic and multi-agent workflows, tool use and function calling design, RAG and retrieval pipelines, prompt and context engineering, autonomous agents with production guardrails and human-in-the-loop review

EXPERIENCE

Embedded UI Engineer · Hach Company · Loveland, CO · Hybrid

Jan 2026 – Present

Contract Frontend Engineer

Apr 2025 – Jan 2026

Promoted from Contract role; bridging embedded systems and frontend UI across multiple enterprise product lines

- Lead frontend architecture for a large-scale TypeScript/Lit monorepo supporting multiple water analysis product lines; established reusable Atomic UI component libraries and MVC patterns adopted across concurrent product teams.
- Engineer real-time device management interfaces integrating live MQTT and HTTP data streams, delivering low-latency UI updates critical for lab and field instrumentation workflows.
- Define and enforce frontend engineering standards through custom ESLint rule sets and comprehensive Jest unit test suites; manage UI submodule dependencies across product lines to ensure stable, coordinated releases.
- Act as the technical bridge between embedded firmware teams and UI, translating hardware data pipelines into intuitive, responsive interfaces for engineers, lab technicians, and enterprise clients.
- Embed Claude Code and GitHub Copilot into daily frontend engineering, connecting MCP servers to Figma and the Atlassian suite for design-to-code handoff and living documentation.

Software Engineer · Philadelphia Scientific, LLC · Montgomeryville, PA

Sept 2022 – Apr 2025

- Led full-stack frontend development of multiple production web applications for Fortune 500 clients including Target, TJX, and Uline, independently owning the full lifecycle from component design through AWS deployment.
- Architected a real-time operations dashboard with interactive data visualizations and live data feeds; built zero-downtime CI/CD pipeline via GitHub Actions with automated deployment to AWS S3.
- Built a touch-optimized React PWA for on-device battery maintenance with a Python/Flask REST API backend; deployed and managed enterprise-wide across major national distribution centers via SureMDM MDM.
- Designed React/TypeScript component libraries with Bootstrap, enforcing consistent UX patterns and accessibility across all client-facing products; provided ongoing production support maintaining uptime for business-critical applications.

Software Engineer · Regulus Group, LLC · Atlantic City, NJ

Jun 2021 – Sept 2022

- Modernized data-input and analytical-output UIs for FAA Surveillance Branch operators and built real-time/historical performance dashboards with dynamic data visualizations, optimizing SQL queries for high-volume analytical workloads on mission-critical government systems.

PROJECTS

PassportIQ · Founder & Full-Stack Engineer, Mattermind · passportiq.app

- Founded and shipped a multi-tenant SaaS platform that helps EU battery manufacturers meet the 2027 EU Digital Product Passport regulation, owning product, full-stack engineering, and deployment end to end.
- Built a React/TypeScript app over a FastAPI/PostgreSQL backend with Stripe subscription billing, GS1 Digital Link QR passports, fleet compliance dashboards, and CSV bulk import, backed by a CI/CD pipeline and 300+ automated tests (pytest + Playwright).

Stack: React · TypeScript · FastAPI · Python · PostgreSQL · Stripe · Playwright · Vercel · Railway

Battery Manager Operations Dashboard · Philadelphia Scientific

- Enterprise real-time ops dashboard with live data visualization and interactive charts; fully automated AWS S3 + GitHub Actions deployment pipeline serving Fortune 500 distribution center operators.

Stack: React · TypeScript · Python · SQL · AWS S3 · GitHub Actions

EDUCATION

B.S. Computer Science · Rowan University, Glassboro, NJ

December 2020